SMOOTH LOCUS, FORMAL SMOOTHNESS OF A VERSAL RING, AND UNOBSTRUCTEDNESS ARE EQUIVALENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original reference 0DQU: sources/stacks-chow/tags.txt, source lines 17199–17199. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0DZR: sources/stacks-chow/tags.txt, source lines 17511–17511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0DZK: sources/stacks-chow/tags.txt, source lines 17505–17505. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DQX: sources/stacks-chow/tags.txt, source lines 17202–17202. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0DQY: sources/stacks-chow/tags.txt, source lines 17203–17203. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07EF: sources/stacks-chow/tags.txt, source lines 9445–9445. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07PM: sources/stacks-chow/tags.txt, source lines 9802–9802. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07NT: sources/stacks-chow/tags.txt, source lines 9773–9773. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:stacks-geometry.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0DQZ: sources/stacks-chow/tags.txt, source lines 17204–17204. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 07VH: sources/stacks-chow/tags.txt, source lines 10007–10007. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0DQU. Let $x_0 : \mathrm{Spec}(k) \to \mathcal{X}$ be a morphism such that $\mathrm{Spec}(k) \to S$ is of finite type with image s. Let A be a versal ring to $\mathcal{X}$ at x_0. The following are equivalent*

1. *x_0 is in the smooth locus of $\mathcal{X} \to S$ (Morphisms of Stacks, Lemma 0DZR),*
2. *$\mathcal{O}_{S,s} \to A$ is formally smooth in the $\mathfrak{m}_A$-adic topology, and*
3. *$\mathcal{F}_{\mathcal{X},k,x_0}$ is unobstructed.*

Proof. The equivalence of (2) and (3) follows immediately from Formal Deformation Theory, Lemma 0DZK.

Note that $\mathcal{O}_{S,s} \to A$ is formally smooth in the $\mathfrak{m}_A$-adic topology if and only if $\mathcal{O}_{S,s} \to A' = A[[t_1, \ldots, t_r]]$ is formally smooth in the $\mathfrak{m}_{A'}$-adic topology. Hence (2) does not depend on the choice of our versal ring by Lemma 0DQX. Next, let l/k be a finite extension and choose $A \to A'$ as in Lemma 0DQY. If $\mathcal{O}_{S,s} \to A$ is formally smooth in the $\mathfrak{m}_A$-adic topology, then $\mathcal{O}_{S,s} \to A'$ is formally smooth in the $\mathfrak{m}_{A'}$-adic topology, see More on Algebra, Lemma 07EF. Conversely, if $\mathcal{O}_{S,s} \to A'$ is formally smooth in the $\mathfrak{m}_{A'}$-adic topology, then $\mathcal{O}_{S,s}^\wedge \to A'$ and $A \to A'$ are regular (More on Algebra, Proposition 07PM) and hence $\mathcal{O}_{S,s}^\wedge \to A$ is regular (More on Algebra, Lemma 07NT), hence $\mathcal{O}_{S,s} \to A$ is formally smooth in the $\mathfrak{m}_A$-adic topology (same lemma as before). Thus the equivalence of (2) and (1) holds for k and x_0 if and only if it holds for l and $x_{0,l}$.

Choose a scheme U and a smooth morphism $U \to \mathcal{X}$ such that $\mathrm{Spec}(k) \times_{\mathcal{X}} U$ is nonempty. Choose a finite extension l/k and a point $w_0 : \mathrm{Spec}(l) \to \mathrm{Spec}(k) \times_{\mathcal{X}} U$. Let $u_0 \in U$ be the image of w_0. We may apply the above to l/k and to $l/\kappa(u_0)$ to see that we can reduce to u_0. Thus we may assume $A = \mathcal{O}_{U,u_0}^\wedge$, see Lemma 0DQZ. Observe that x_0 is in the smooth locus of $\mathcal{X} \to S$ if and only if u_0 is in the smooth locus of $U \to S$, see for example Morphisms of Stacks, Lemma 0DZR. Thus the equivalence of (1) and (2) follows from More on Algebra, Lemma 07VH. □